Annotated Summary of the Reference Papers

Datasets, baselines, and results on the Nazari benchmark

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This document summarises every paper in `ref/`. For each paper it states (i) what the paper does, (ii) on which datasets it was evaluated (Nazari-style uniform instances, Uchoa Set X / CVRPLIB, Queiroga XML100, Queiroga Set XL, TSPLIB, ...), and (iii) which other solvers it benchmarked. Section 6 gives a one-glance overview table, and Section 7 collects the published results on the Nazari benchmark for $n = 50$ and $n = 100$ customers.

1 The four benchmark families

Nazari benchmark (uniform random instances). Introduced by Nazari et al. [19] and adopted by essentially all neural CVRP papers. Customer and depot coordinates are drawn uniformly from the unit square, demands uniformly from $\{1, \dots, 9\}$, and the vehicle capacity is 30, 40 and 50 for $n = 20$, 50 and 100 customers respectively. Kool et al. [13] fixed the now-standard test protocol of 10,000 held-out instances per size and reported average tour length, which makes numbers across papers directly comparable (Table 2).

Uchoa Set X (and derivatives XE, CVRPLIB). Uchoa et al. [23] proposed 100 instances with 100–1,000 customers covering a wide range of depot positions, customer layouts (random / clustered / mixed), demand distributions and route lengths; it is the standard OR benchmark, hosted on CVRPLIB together with the older classical sets (A, B, E, F, M, P, Golden, ...). The paper evaluates the state-of-the-art heuristics ILS-SP (Subramanian et al.) and UHGS (Vidal et al.) plus a Branch-Cut-and-Price exact code on the new set. Hottung André and Tierney Kevin [10] later derived the *XE* groups (17 groups of 20 instances, 100–297 customers) by re-sampling instances with Set-X characteristics so that learning methods can be trained per distribution.

Queiroga XML100 ("10,000 optimal solutions"). Queiroga et al. [21] (note: *Queiroga*, often misspelled Quiroga) generated 10,000 heterogeneous 100-customer instances with the Set-X generator and solved almost all of them to *proven optimality* with a Branch-Cut-and-Price solver. The package includes the instance generator (for producing unlimited training data) and is explicitly aimed at machine-learning-based heuristics, enabling absolute optimality-gap evaluation.

Queiroga Set XL. Queiroga et al. [22] extend the X methodology to 100 large-scale instances with 1,000–10,000 customers. To provide initial best-known solutions they ran eight solvers: AILS-II, FILO, FILO2, KGLS-XXL, **HGS-CVRP**, SISRs, LKH-3 and OR-Tools. AILS-II produced 93 of the 100 initial BKS; HGS-CVRP remains the best method on XML100 and on the smaller half of Set X, while AILS-II and SISRs scale better to the XL regime. A community BKS challenge is hosted on CVRPLIB.

2 Classical solvers and benchmark papers

LKH-3 [8]. Extension of the Lin–Kernighan–Helsgaun TSP solver to constrained routing problems by transforming them into standard TSPs and penalising constraint violations. Tested on roughly 10,000 benchmark instances from the literature covering ∼30 problem classes; for the CVRP alone, 509 CVRPLIB instances (LKH-3 matched the BKS on 327 of them). It is the most common "strong classical baseline" in the neural-CVRP literature.
Benchmarks: CVRPLIB and many other VRP-variant libraries (no Nazari-style instances in the report itself).
Compared against: best-known solutions from the literature.

HGS-CVRP [24]. Open-source specialisation of the Hybrid Genetic Search to the CVRP, adding the SWAP* neighborhood. On Set X it dominates all competitors (average gap ≈ 0.1%), followed by SISR (0.19%), FILO, HILS, KGLS, LKH-3 (1.00%) and OR-Tools (4.01%). Currently the reference heuristic for the CVRP; several neural papers use it as the optimality proxy.
Benchmarks: Uchoa Set X (100–1,000 customers).
Compared against: OR-Tools, LKH-3, HILS, KGLS, SISR, FILO, HGS-2012.

Set X paper [23], XML100 [21], Set XL [22]. See Section 1.

3 Learning to construct solutions

Pointer Networks [25]. Sequence-to-sequence model with an attention-based "pointer" over input positions, trained by supervised learning. Solves convex hull, Delaunay triangulation and planar TSP with 5–50 cities. No CVRP; historical starting point of the field.
Benchmarks: Self-generated uniform TSP instances.
Compared against: exact/approximate TSP algorithms (A1–A3 heuristics).

Neural Combinatorial Optimization with RL [1]. Trains a pointer network with actor-critic REINFORCE instead of supervision; introduces *active search* (instance-wise fine-tuning at test time). Near-optimal on uniform TSP20–100 and 0–1 Knapsack. No CVRP.
Benchmarks: Uniform random TSP20/50/100, Knapsack.
Compared against: Christofides, OR-Tools local search, supervised pointer network.

Nazari et al. [19]. First deep-RL model for the CVRP: replaces the RNN encoder of the pointer network with permutation-invariant element-wise embeddings so that the dynamic state (remaining demands) can be re-embedded after each decoding step; trained with policy gradient, greedy or beam-search decoding. Defines the Nazari benchmark (VRP10/20/50/100, see Section 1). Results on VRP50/100 are reported as win rates rather than tour lengths; the lengths usually attributed to this method (11.15 / 16.96 for RL-BS(10)) were tabulated by Kool et al. [13].
Benchmarks: Nazari uniform instances (VRP10–VRP100), 1,000 test instances.
Compared against: Clarke–Wright savings, Sweep heuristic, OR-Tools, exact MIP (small sizes).

Attention Model (AM) [13]. Transformer encoder + attention decoder trained with REINFORCE and a greedy-rollout baseline; one architecture for TSP, CVRP, SDVRP, OP and PCTSP. Established the 10k-instance evaluation protocol. CVRP: 10.98/16.80 (greedy) and 10.62/16.23 (1280 samples) for n=50/100.
Benchmarks: Nazari uniform instances (TSP/CVRP/SDVRP/OP/PCTSP, $n \leq 100$).
Compared against: Gurobi, Concorde, LKH-3, OR-Tools, Clarke–Wright, Sweep, Nazari et al., NeuRewriter-era baselines.

POMO [14]. Exploits solution symmetries: REINFORCE with a shared baseline over N rollouts starting from every possible first node, plus $\times 8$ instance augmentation at inference. CVRP100 in 0.32% of LKH-3 within minutes (10.42 / 15.73 for n=50/100).

Benchmarks: Nazari uniform instances (TSP, CVRP, Knapsack).
Compared against: LKH-3, OR-Tools, AM, NeuRewriter, NLNS, L2I.

Learning Collaborative Policies (LCP) [12]. Two-policy scheme: a "seeder" generates diverse candidate solutions (entropy-regularised AM) and a "reviser" repairs partial segments; applicable on top of AM. AM+LCP reaches 10.52 / 15.98 on CVRP50/100.

Benchmarks: Nazari uniform instances (TSP, PCTSP, CVRP; TSP up to n=500).
Compared against: LKH-3, OR-Tools, AM (sampling), NLNS, Nazari RL.

Efficient Active Search (EAS) [9]. Revives Bello et al.'s active search but updates only a small subset of parameters per instance (added embeddings, an extra layer, or a policy lookup table) on top of a trained POMO model. Surpasses LKH-3 on uniform CVRP100 (15.63 vs. 15.65) and generalises to $n = 125/150/200$; also evaluated on the XE groups derived from Set X, where EAS beats NLNS and LKH-3 on 8 of 9 sets.

Benchmarks: Nazari/Kool uniform instances (TSP/CVRP n=100–200); XE instance groups based on Set X [10].
Compared against: LKH-3, UHGS (on XE), Concorde, POMO (greedy/sampling), original active search, DPDP, NLNS, DACT, GNN-guided methods.

Simulation-Guided Beam Search (SGBS) [4]. Inference method: beam search whose expansions are scored by greedy POMO rollouts (simulations), optionally interleaved with EAS updates (SGBS+EAS). CVRP100: 15.659 (SGBS) and 15.594/15.580 (SGBS+EAS short/long); roughly halves the previous neural gap to HGS.

Benchmarks: Uniform instances: TSP100/150/200, CVRP100/150/200, FFSP20/50/100.
Compared against: Concorde, LKH-3, HGS, DACT, NLNS, DPDP, POMO, EAS, MatNet (FFSP).

BQ-NCO [7]. Reformulates constructive NCO as a bisimulation-quotiented MDP so that the policy conditions only on the *remaining* sub-problem; trained by imitation of near-optimal solutions (LKH for CVRP). Strong out-of-distribution generalisation: trained on $n = 100$, tested up to $n = 1{,}000$ (CVRP gaps vs. LKH-3: 0.61% at n=100 with beam 16) and on all Euclidean CVRPLIB instances up to 4,461 nodes.

Benchmarks: Uniform instances $n \in \{100, 200, 500, 1000\}$ (TSP, CVRP, OP, KP); TSPLIB; CVRPLIB (incl. Set X).
Compared against: LKH-3, HGS, Concorde, OR-Tools, AM, MDAM, POMO, SGBS, EAS, LEHD.

LEHD [16]. Light-encoder/heavy-decoder architecture trained by supervised learning on HGS solutions of uniform CVRP100; the heavy decoder re-attends to all remaining nodes at each step, which transfers well to larger sizes. With Random Re-Construct (RRC) iterations it outperforms LKH-3 on uniform CVRP200–500 (gaps are reported relative to LKH-3, e.g. -0.24% at n=200 with RRC 500). Also tested on TSPLIB and CVRPLIB sets A, B, E, F, M, P and X.

Benchmarks: Uniform instances $n \in \{100, 200, 500, 1000\}$; TSPLIB; CVRPLIB sets A, B, E, F, M, P, X.
Compared against: LKH-3, HGS, Concorde, OR-Tools, MDAM, POMO, SGBS, EAS, BQ-NCO, Att-GCN+MCTS.

Self-Improvement / Gumbeldore [20]. Replaces both expert data and policy gradients: in each epoch the model samples solutions without replacement (round-wise stochastic beam search with an advantage-based policy update), then imitates its own best solutions. Built on the BQ architecture; matches supervised BQ on TSP/CVRP (CVRP100 gap to HGS: 1.72% with beam 16 vs. 1.22% for supervised BQ) and sets state of the art on JSSP.

Benchmarks: Uniform instances $n \in \{100, 200, 500, 1000\}$ (same test sets as LEHD/BQ-NCO); JSSP benchmarks.
Compared against: OR-Tools, AM, MDAM, POMO, SGBS, BQ (supervised), LEHD.

4 Learning to improve / learning-guided search

NeuRewriter [3]. Learns a region-picking and a rule-picking policy that iteratively rewrite a feasible solution; trained with actor-critic on the improvement per rewriting step. Applications: expression simplification, online job scheduling, CVRP (6.16 / 10.51 / 16.10 on CVRP20/50/100).

Benchmarks: Nazari uniform instances (CVRP20/50/100).
Compared against: OR-Tools, Clarke–Wright, Sweep, Nazari RL, AM.

L2I – Learn to Improve [15]. Iterative framework alternating learned improvement operators (RL-selected intra/inter-route moves) with rule-based perturbation operators; an ensemble of six policies with different history horizons. First learned method to beat LKH-3 on the Nazari benchmark: 6.12 / 10.35 / 15.57 on CVRP20/50/100 (but with hours-long search budgets).

Benchmarks: Nazari uniform instances (CVRP20/50/100).
Compared against: LKH-3, OR-Tools, Nazari RL, AM, NeuRewriter.

Learning Improvement Heuristics [27]. Self-attention policy that selects the node pair for a pairwise local-search operator (2-opt, swap, reinsertion) inside a fixed-step improvement loop; RL with a best-so-far-based reward. TSP and CVRP: 10.45 / 16.03 on CVRP50/100 with T=5,000 steps; also generalises to TSPLIB and CVRPLIB instances.

Benchmarks: Nazari uniform instances (TSP/CVRP 20/50/100); TSPLIB; CVRPLIB.
Compared against: LKH-3, OR-Tools, AM (sampling), NeuRewriter.

Learning 2-opt Heuristics [6]. Policy-gradient method (with GAE) that learns to pick 2-opt moves for the TSP using a pointing-attention mechanism; improves even from random initial tours. TSP only (TSP20/50/100 uniform); relevant here as the TSP analogue of learned local search.

Benchmarks: Uniform TSP20/50/100.
Compared against: Concorde/exact, AM, Wu et al., classical 2-opt.

NLNS – Neural Large Neighborhood Search [10]. LNS in which the *repair* operator is a learned attention model (destroy operators are handcrafted); batch search for many instances in parallel, single-instance search with simulated-annealing acceptance. Batch search: 10.54 / 15.99 on the Nazari sets. Single instance search on the 17 XE groups (Set-X characteristics, 100–297 customers) reaches parity with LKH-3, behind UHGS.

Benchmarks: Nazari uniform instances (CVRP/SDVRP 20/50/100); XE groups derived from Set X.
Compared against: AM (sampling), Nazari RL-BS, UHGS, LKH-3, handcrafted-repair LNS.

DACT – Dual-Aspect Collaborative Transformer [17]. Improvement model encoding node features and positional (sequence) features in two separate embedding streams with cyclic positional encoding; PPO with curriculum learning selects 2-opt-style moves. CVRP: 10.39 / 15.71 with $\times 6$ augmentation; generalises to TSPLIB/CVRPLIB ($\leq$ 200 customers) better than POMO.

Benchmarks: Nazari uniform instances (TSP/CVRP 20/50/100); TSPLIB and CVRPLIB ($\leq$ 200).
Compared against: LKH-3, OR-Tools, AM, MDAM, POMO ($\times 8$), Wu et al., NeuRewriter, NLNS, DPDP, CVAE-Opt.

NeuOpt [18]. Learning-to-search solver performing flexible k-opt via a recurrent dual-stream decoder; the GIRE scheme drops feasibility masking and lets the policy explore infeasible regions with reward shaping (capacity constraint of CVRP), plus dynamic data augmentation (D2A). First learned local-search method to surpass LKH-3 on the Nazari CVRP sets: 10.367 / 15.579 at the largest budget, essentially matching HGS (10.366 / 15.563).

Benchmarks: Nazari uniform instances (TSP/CVRP 20/50/100); TSPLIB; CVRPLIB Set X ($\leq$ 200 customers).
Compared against: HGS, LKH-3, Concorde, POMO, POMO+EAS(+SGBS), AM+LCP, Sym-NCO, DACT, Wu et al., NLNS, NCE, DPDP, CVAE-Opt.

Neural Simulated Annealing [5]. Frames the SA proposal distribution as an RL policy (small equivariant networks) trained with PPO or evolutionary strategies; the Metropolis criterion and temperature schedule stay classical. Evaluated on Rosenbrock, Knapsack, Bin Packing and TSP (up to $n \approx 200$ with strong size generalisation). No CVRP – conceptually the closest relative of LG-SA.

Benchmarks: Rosenbrock's function, Knapsack, Bin Packing, uniform TSP.
Compared against: vanilla SA, greedy/random baselines, learned TSP solvers.

GNN-Guided Local Search [11]. GNN predicts the "regret" of each edge; guided local search penalises high-regret edges. TSP only: reduces the TSP100 gap of learned methods from 1.5% to 0.7%.

Benchmarks: Uniform TSP20–100 (Kool test sets), TSPLIB.
Compared against: Concorde, LKH, AM, GCN-based predictors, 2-opt/GLS variants.

RHGA – Reinforced Hybrid Genetic Algorithm [28]. Combines the EAX genetic algorithm with LKH local search for the *TSP*; a Q-learning mechanism (from their earlier VSR-LKH work) adapts the candidate-edge selection. Evaluated on 138 TSPLIB/national instances with 1,000–85,900 cities.

Benchmarks: TSPLIB and large TSP benchmarks.
Compared against: EAX-GA, LKH, VSR-LKH.

RFTHGS – RL-finetuned LLM for HGS [29]. Fine-tunes a small code LLM with a curriculum-based RL reward to generate *crossover operators* for HGS-CVRP that outperform the expert-designed OX crossover. Evaluated on CVRPLIB Set X instances from small to 1,000 nodes.

Benchmarks: CVRPLIB Set X (up to 1,000 customers).
Compared against: HGS with expert operators, POMO, MTPOMO, RF-POMO and other neural solvers, prompt-based LLM heuristic design (EoH etc.), GPT-4o / GPT-4o-mini.

5 Surveys

ML for Combinatorial Optimization [2]. Methodological tour d'horizon: taxonomy of learning alongside/instead of OR algorithms (learning to configure, learning in branching, end-to-end heuristics), generalisation and evaluation methodology. No experiments.

NCO for VRPs: Comprehensive Survey [26]. Taxonomy of neural VRP solvers into Learning-to-Construct, Learning-to-Improve, Learning-to-Predict-Once and Learning-to-Predict-Multiplicity; discusses generalisation, scale, variants and OR-vs-ML comparability, and compares representative solvers across scales on uniform instances and TSPLIB/CVRPLIB; maintains a live repository.

6 Overview: who tests on what

Paper	Test datasets	Solvers benchmarked
Nazari et al. [2018]	Nazari uniform VRP10–100	CW savings, Sweep, OR-Tools, MIP
AM [2018]	Nazari uniform ($n \leq 100$)	Gurobi, Concorde, LKH-3, OR-Tools, Nazari RL
NeuRewriter [2019]	Nazari uniform	OR-Tools, Nazari RL, AM
L2I [2020]	Nazari uniform	LKH-3, OR-Tools, Nazari RL, AM, NeuRewriter
Wu et al. [2022]	Nazari uniform; TSPLIB; CVRPLIB	LKH-3, OR-Tools, AM, NeuRewriter
NLNS [2020]	Nazari uniform; XE (Set-X style)	AM, Nazari RL, UHGS, LKH-3
POMO [2020]	Nazari uniform	LKH-3, OR-Tools, AM, NeuRewriter, NLNS, L2I
LCP [2021]	Nazari uniform	LKH-3, OR-Tools, AM, NLNS
DACT [2022]	Nazari uniform; TSPLIB/CVRPLIB ≤ 200	LKH-3, OR-Tools, AM, MDAM, POMO, Wu et al., NLNS, DPDP
EAS [2021]	Kool uniform n=100–200; XE	LKH-3, UHGS, POMO, DPDP, NLNS, DACT, active search
SGBS [2022]	Uniform CVRP100–200, TSP, FFSP	LKH-3, HGS, POMO, EAS, DACT, NLNS, DPDP
NeuOpt [2023]	Nazari uniform; TSPLIB; Set X $\leq$ 200	HGS, LKH-3, POMO(+EAS,+SGBS), DACT, NLNS, ...
BQ-NCO [2023]	Uniform n=100–1000; TSPLIB; CVRPLIB	LKH-3, HGS, OR-Tools, POMO, SGBS, EAS, LEHD
LEHD [2023]	Uniform n=100–1000; TSPLIB; CVRPLIB A/B/E/F/M/P/X	LKH-3, HGS, OR-Tools, POMO, SGBS, EAS, BQ
Self-Improvement [2024]	Uniform n=100–1000; JSSP	OR-Tools, AM, MDAM, POMO, SGBS, BQ, LEHD
RFTHGS [2025]	CVRPLIB Set X ($\leq$ 1000)	HGS, POMO variants, LLM-designed heuristics, GPT-4o
Neural SA [2022]	Rosenbrock, KP, BPP, TSP	vanilla SA, off-the-shelf solvers
Set X [2017]	Set X (introduced)	ILS-SP, UHGS, Branch-Cut-and-Price
XML100 [2022]	XML100 (introduced)	Branch-Cut-and-Price (optimal solutions)
Set XL [2026]	Set XL (introduced); X; XML100	AILS-II, FILO(2), KGLS-XXL, HGS-CVRP, SISRs, LKH-3, OR-Tools
HGS-CVRP [2022]	Set X	OR-Tools, LKH-3, HILS, KGLS, SISR, FILO, HGS-2012
LKH-3 [2017]	CVRPLIB + $\sim$30 VRP-variant libraries	best-known solutions

TSP-only or non-CVRP papers (Pointer Networks, Bello et al., da Costa et al., GNN-GLS, RHGA, Neural SA) and the two surveys are omitted from the CVRP tables.

7 Results on the Nazari benchmark ($n = 50$ and $n = 100$)

Table 2 collects the average tour lengths on the Nazari-distribution test sets (uniform coordinates, demands $\in \{1, \ldots, 9\}$, capacity 40/50). Most values are on Kool's 10,000-instance test sets. For each size we report two times: "10k" is the *total wall-clock time to solve the full 10,000-instance test set* as reported in the source of the corresponding row, and "/inst." is the amortized time per instance (total divided by the number of instances). Beware that neural methods solve the test set in large GPU batches, so the amortized per-instance time can be much lower than the latency of solving one instance alone; and since hardware, batching and implementation language differ across papers, times are indicative only: they rank search-budget regimes (milliseconds vs. minutes per instance), not implementations.

Table 2: Average tour length and solving time on Nazari-distribution instances, CVRP50 and CVRP100. "10k" = total time for the 10,000-instance test set; "/inst." = amortized time per instance.

	CVRP50			CVRP100			
Method	Obj.	10k	/inst.	Obj.	10k	/inst.	Source
Classical solvers							
HGS-CVRP	10.366	1.2d	10s	15.563	2.5d	22s	[18]
LKH-3	10.38	7h	2.5s	15.65	13h	4.7s	[13]
OR-Tools	11.22	12m	0.07s	17.14	1h	0.36s	[14]
Learning to construct							
Nazari RL, BS(10)	11.15	n/r	n/r	16.96	n/r	n/r	[13]
AM, greedy	10.98	3s	0.3ms	16.80	8s	0.8ms	[13]
AM, 1280 samples	10.62	28m	0.17s	16.23	2h	0.72s	[13]
POMO, ×8 augm.	10.42	26s	2.6ms	15.73	2m	12ms	[14]
AM + LCP $\{2560, 1\}$	10.52	52m	0.31s	16.00	2.1h	0.76s	[18]
POMO + EAS-Emb	10.379	3.1h	1.1s	15.610	16h	5.8s	[18]
POMO + EAS + SGBS (long)	—			15.580	30h	11s	[4]
DPDP (heatmap + DP)	—			15.63	23h	8.3s	[9]
Learning to improve / search							
NeuRewriter	10.51	≈55m‡	0.33s	16.10	≈85m‡	0.51s	[3]
L2I	10.35†	≈118d§	≈17m	15.57†	≈167d§	≈24m	[15]
Wu et al., T=5k	10.45	4h	1.4s	16.03	5h	1.8s	[27]
NLNS, batch search	10.54	24m	0.15s	15.99	62m	0.37s	[10]
DACT, ×6 augm.	10.39	1.5h	0.54s	15.71	4.5h	1.6s	[17]
NeuOpt-GIRE (D2A=5, T=40k)	10.367	1.6d	14s	15.579	3.8d	33s	[18]

n/r = not reported. † authors' own test set of the same distribution (small differences ∼0.01 are not meaningful). ‡ measured on 2,000 instances (11m / 17m, as re-reported by Ma et al. [17]); the 10k column scales this by 5. § L2I is the reverse case: its original paper reports time *per single instance* (read from its runtime figure; ≈17m / ≈24m), and the 10k column is the extrapolation — Ma et al. [17] estimate ≈167 days for the full 10,000-instance CVRP100 set.

Gap-only results. Recent generalisation-focused papers report optimality gaps instead of lengths, and only for $n \geq 100$: BQ-NCO reaches 0.61% above LKH-3 on uniform CVRP100 (beam 16) [7]; LEHD reaches 0.27% with 100 RRC iterations and beats LKH-3 with larger budgets [16]; the self-improvement variant of BQ reaches 1.72% above HGS on CVRP100 (beam 16) versus 1.22% for supervised BQ [20].

Reading the table. Three observations recur across these papers. (i) Construction methods (POMO family) dominate the speed–quality trade-off at small budgets, while learned local search (DACT, NeuOpt) and hybrid search (EAS, SGBS) close the gap to LKH-3/HGS at large budgets. (ii) Every method above is trained and tested on the *same* uniform distribution; performance degrades sharply on Set X-style structured instances, which is exactly what the XE/XML100/XL benchmarks were designed to expose. (iii) L2I's 15.57 and NeuOpt's 15.579 on CVRP100 are the strongest published learned-search results on this benchmark, but L2I needs tens of minutes *per instance* and NeuOpt several days for the full test set, whereas HGS attains 15.563 as a mature standalone heuristic.

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